

Model	Dimension	λ	L0 (Sparsity)	Alive Feature Map	FVU	Test Acc (%)	Inner Product
Baseline (CNN)	512	-	N/A	512	N/A	90.31 $\pm$ 0.38	N/A
CNN Proxy	600	50	4.9 $\pm$ 0.2	587.0 $\pm$ 4.8	0.3444 $\pm$ 0.0209	89.24 $\pm$ 0.50	0.160 $\pm$ 0.023
Effect of Dictionary Expansion (Fixed $\lambda=800$)							
Gated SAE	1024	800	2.7 $\pm$ 0.2	1020.8 $\pm$ 3.1	0.7295 $\pm$ 0.0155	88.48 $\pm$ 1.27	0.104 $\pm$ 0.018
Gated SAE	2048	800	4.2 $\pm$ 0.2	2034.6 $\pm$ 4.2	0.5306 $\pm$ 0.0151	89.63 $\pm$ 0.71	0.089 $\pm$ 0.008
Gated SAE	4096	800	7.6 $\pm$ 0.2	3969.5 $\pm$ 39.4	0.3976 $\pm$ 0.0135	89.52 $\pm$ 0.51	0.082 $\pm$ 0.005
Gated SAE	8192	800	9.8 $\pm$ 0.6	3733.4 $\pm$ 511.5	0.2953 $\pm$ 0.0145	89.91 $\pm$ 0.47	0.104 $\pm$ 0.012
Effect of Sparsity Penalty (Fixed Dim=4096)							
Gated SAE	4096	200	13.2 $\pm$ 0.2	3850.1 $\pm$ 38.3	0.2414 $\pm$ 0.0140	89.52 $\pm$ 0.51	0.099 $\pm$ 0.007
Gated SAE	4096	800	7.6 $\pm$ 0.2	3969.5 $\pm$ 39.4	0.3976 $\pm$ 0.0135	89.88 $\pm$ 0.59	0.082 $\pm$ 0.005
Gated SAE	4096	3200	2.9 $\pm$ 0.2	3866.2 $\pm$ 89.7	0.7190 $\pm$ 0.0149	88.23 $\pm$ 1.30	0.052 $\pm$ 0.005